META GLOBAL AI DEVELOPER HACKATHON · PUSH THE FRONTIER

AI can recognise what it sees.

But can it prove what's true?

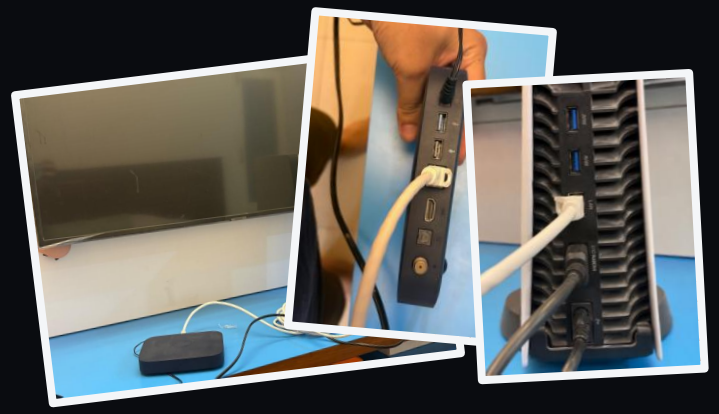
Introducing ProofCam.



Photo: the author's PlayStation 5. Labels are what a live ProofCam inspection recorded (Muse Spark 1.3 + SAM 3.1).

THE PROBLEM

The physical world runs on assumptions.



INSTALLATION CHECKLIST

- Router / set-top box installed
- Console cabled
- TV connected
- Signed off

Photos are collected.

Pictures pile up after the work is done, with no link to what they prove.

Checklists are completed.

Ticked by the person who did the work.

Requirements are assumed.

No one can show which photo proves which connection.

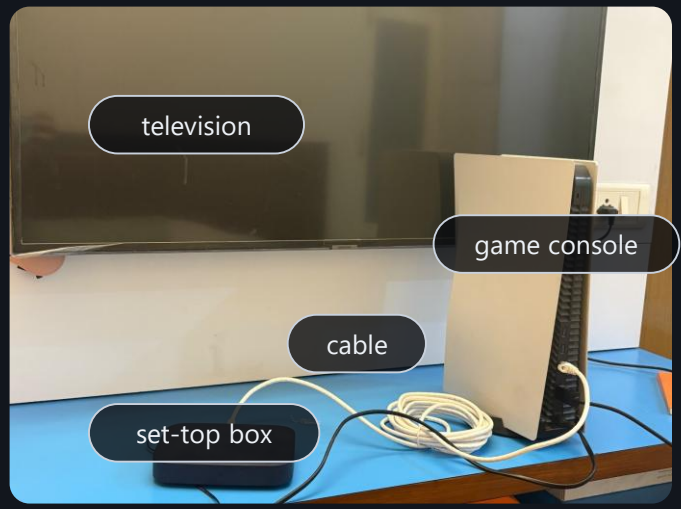
A cable is visible in a photo ≠ **The required connection is proven**

THE INSIGHT

Seeing is not verifying.

TRADITIONAL IMAGE AI

"What objects are in this photograph?"



Illustrative labels on a real photo

OUTPUT

Labels.

No requirement checked. No gap found.

→

PROOFCAM

"What evidence establishes that this requirement is satisfied?"

`R1 · set-top box RJ45 → PS5 LAN · Ethernet`

✓ Set-top box RJ45 end	photo 2 · Muse + SAM
✓ PS5 LAN end	photo 4 · Muse + SAM
? Same cable at both ends	in no photo · gap stays open

OUTPUT

Evidence, the open gap, and the next photo to take.

From passive recognition to active evidence investigation.

THE PRODUCT

Meet ProofCam.

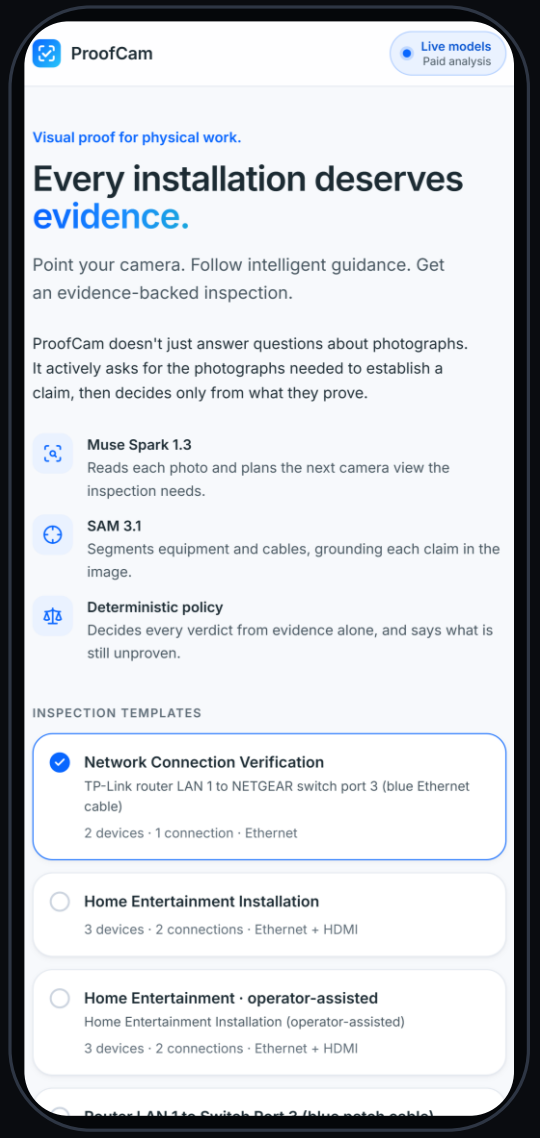
An AI agent for evidence-driven physical inspections.

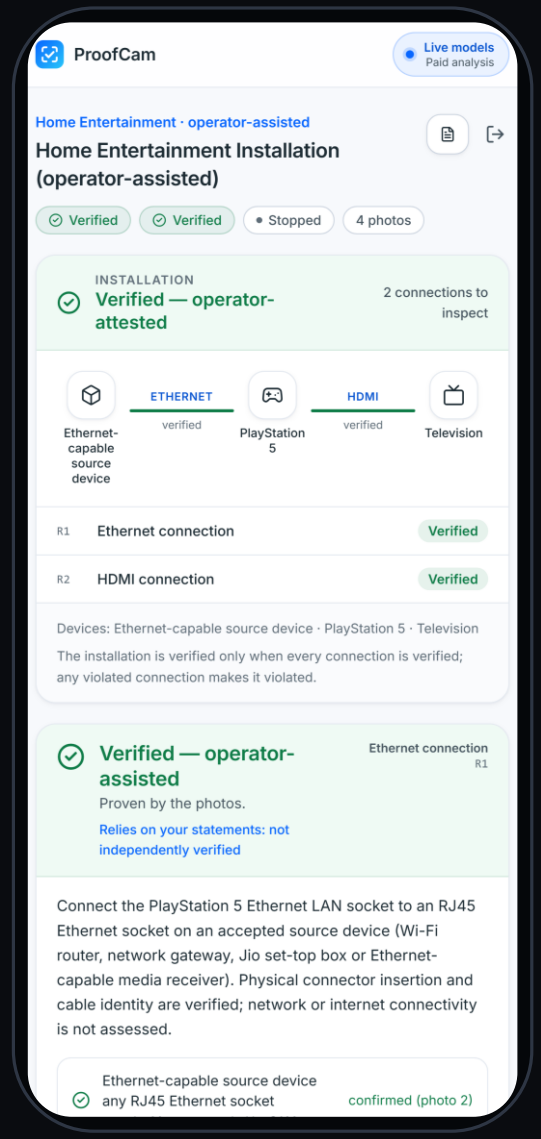
Inspect
Photograph the installation with any phone browser.

Reason
Muse Spark reads ports, plugs and labels; SAM 3.1 grounds them.

Request
It asks for the one view that closes the open gap.

Verify
A deterministic policy decides, with an audit trail.





Real app screens (installable PWA) · start screen · completed real inspection

▶▶ PROCESS ACCELERATED

▶

INSERT RECORDING A

VIOLATED INSPECTION

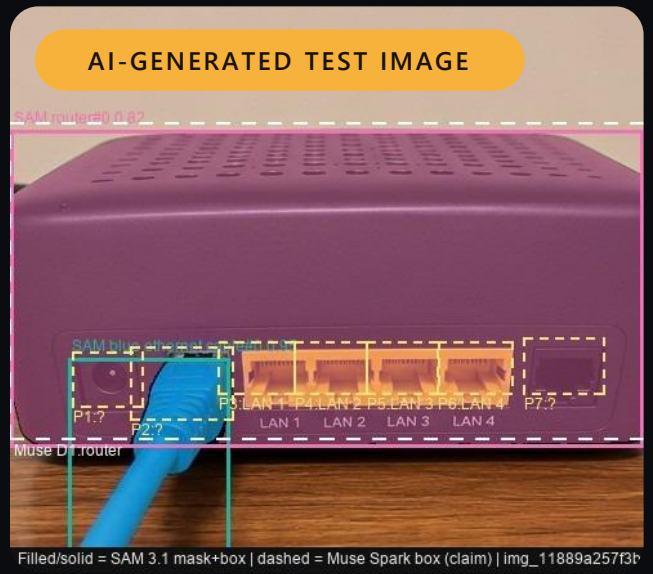
iPhone screen recording · portrait
≈20 s · router LAN 1 → switch port 3
Replace this frame: see speaker notes

LIVE DEMO · RECORDING A

It doesn't approve everything.

REQUIREMENT

The blue Ethernet cable must connect Router LAN 1 to Switch Port 3.



LAN 1 empty

plug unlabelled socket

"A model's observation is not enough. The requirement must be satisfied."

Violated R1

Clearly observed not to meet the requirement.

The blue Ethernet cable must connect Router LAN 1 to Switch Port 3.

Router LAN 1	clearly wrong
Switch 3	not yet established
Same cable at both ends	not yet established

NOT OBSERVED (DOES NOT CHANGE THIS VERDICT)

Device not identified. The Switch has not been identified with certainty in any photo.

Clearly observed: [photo 3] Router port P3 'LAN 1' [legible] (empty, plug=n/a, visibility=clear) → target port is empty.

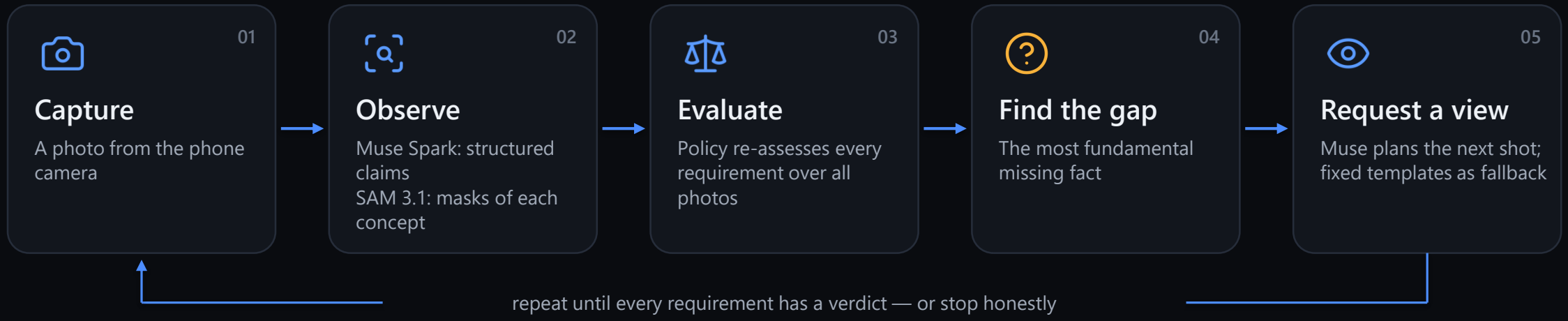
Evidence (1 item) ›

Why (full explanation) ›

Test photos are AI-generated (synthetic) and disclosed · analysed live by Meta Muse Spark 1.3 + SAM 3.1 · 3 photos · 5 model calls

HOW IT WORKS · THE ACTIVE INSPECTION LOOP

ProofCam knows what it still needs to see.



FROM THE REAL INSPECTION · REQUEST nv1 · PLANNED BY MUSE SPARK 1.3

Missing: the set-top box's connection ports have not been seen.

"Walk around to behind the blue table and photograph the rear panel of the dark set-top box on the left…"

Photo 2 → RJ45 plug seen by Muse, corroborated by SAM ✓

FROM THE REAL INSPECTION · REQUEST nv5 · FIXED TEMPLATE

Missing: both HDMI ends show a plug, but no photo shows one cable joining them.

"Step back so one photo shows the hdmi cable run between both endpoints, from one device to the other…"

No such photo was taken → the operator declared it instead, and the verdict says so.

META AI MODELS

Two Meta models. One verification system.

MUSE SPARK 1.3 · REASONS

Reads each photo into structured claims (devices, ports, labels, plugs, cables, certainty) and plans the next camera view for the open gap.

`structured JSON output · vision · next-view planning`

SAM 3.1 · GROUNDS

Segments each named cable and device concept on the same image. A plug claim counts only where a real mask reaches that port.

`concept segmentation · independent spatial check`

DETERMINISTIC POLICY · VERIFIES

Fuses every photo against the specification. It alone sets VERIFIED, VIOLATED or INSUFFICIENT EVIDENCE, with the provenance of each claim.

`no model output is ever a verdict`

Not one ‘looks connected’ prompt: 22 model calls across 4 photos, each claim checked.

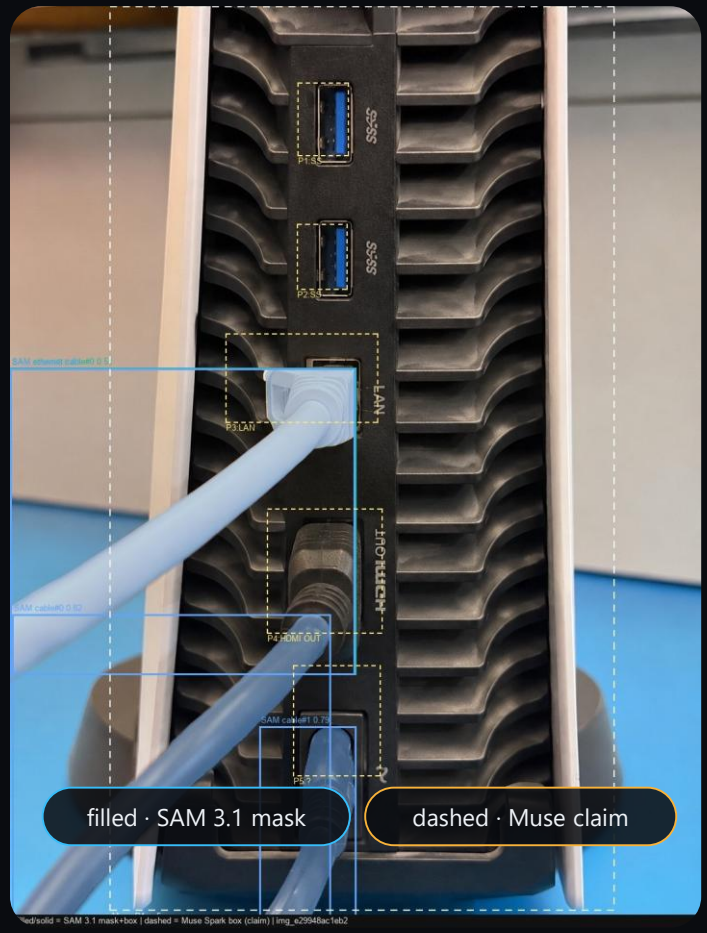


Photo 4, stored overlay · LAN: Muse + SAM ✓ · HDMI OUT: Muse only

▶▶ PROCESS ACCELERATED

INSERT
RECORDING B

REAL EQUIPMENT · 4 PHOTOS

iPhone screen recording · portrait
≈25 s · photos 1→4 and results
Replace this frame: see speaker notes

LIVE DEMO · RECORDING B

Watch the evidence accumulate.



PHOTO 1 · OVERVIEW

TV, PlayStation 5 and Jio set-top box identified

set-top box ports not visible → first next-view request



PHOTO 2 · REQUESTED VIEW

Set-top box rear: RJ45 plug clearly inserted

Muse ✓ · SAM ✓ (generic 'cable' mask) · end established



PHOTO 3 · REQUESTED VIEW

TV panel: plug in HDMI 1 (ARC)

Muse ✓ · SAM ✓ (generic 'cable' mask) · end established



PHOTO 4 · REQUESTED VIEW

PS5 rear: LAN plug and HDMI OUT plug

LAN: Muse ✓ SAM ✓ · HDMI OUT: Muse only (SAM found no HDMI cable)

Real equipment · 4 photos · 22 live Meta model calls (8 Muse Spark 1.3, 14 SAM 3.1) · model waits sped up in the recording

ASSURANCE · HUMAN IN THE LOOP

When the camera can't prove it, ProofCam says so.

ESTABLISHED FROM PHOTOS

Muse observed · SAM corroborated · policy accepted

Set-top box RJ45 plug
photo 2

PlayStation 5 LAN plug
photo 4

TV HDMI input plug
photo 3

CONFIRMED BY THE OPERATOR

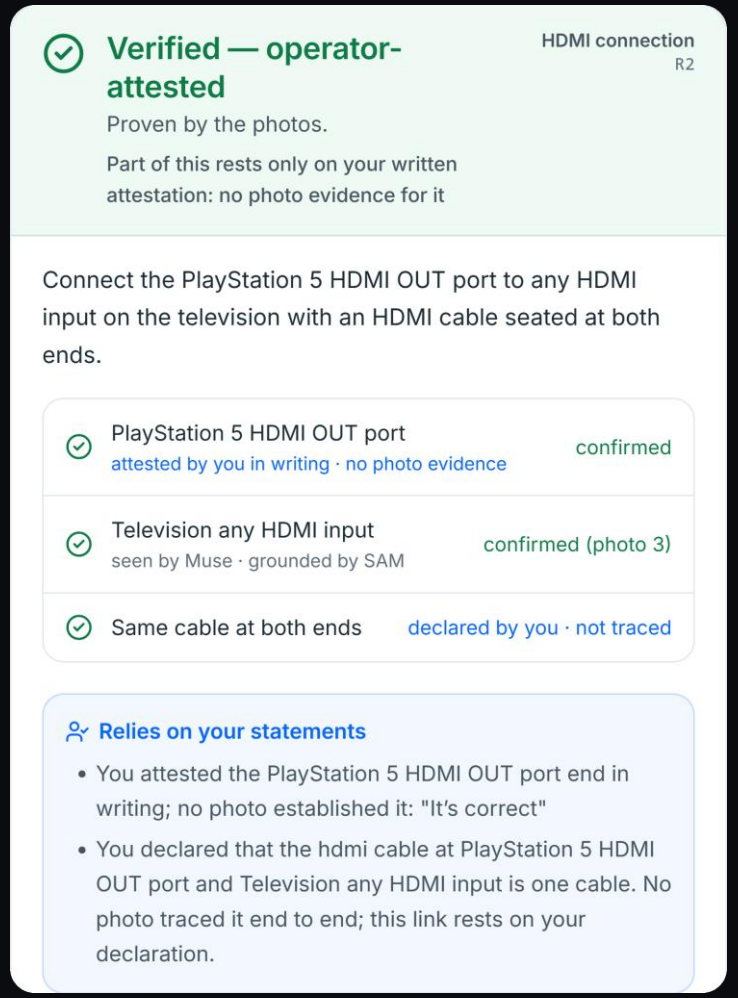
Recorded verbatim · time-stamped · labelled in the verdict

Ethernet run is one cable
declared · no photo traced it

HDMI run is one cable
declared · no photo traced it

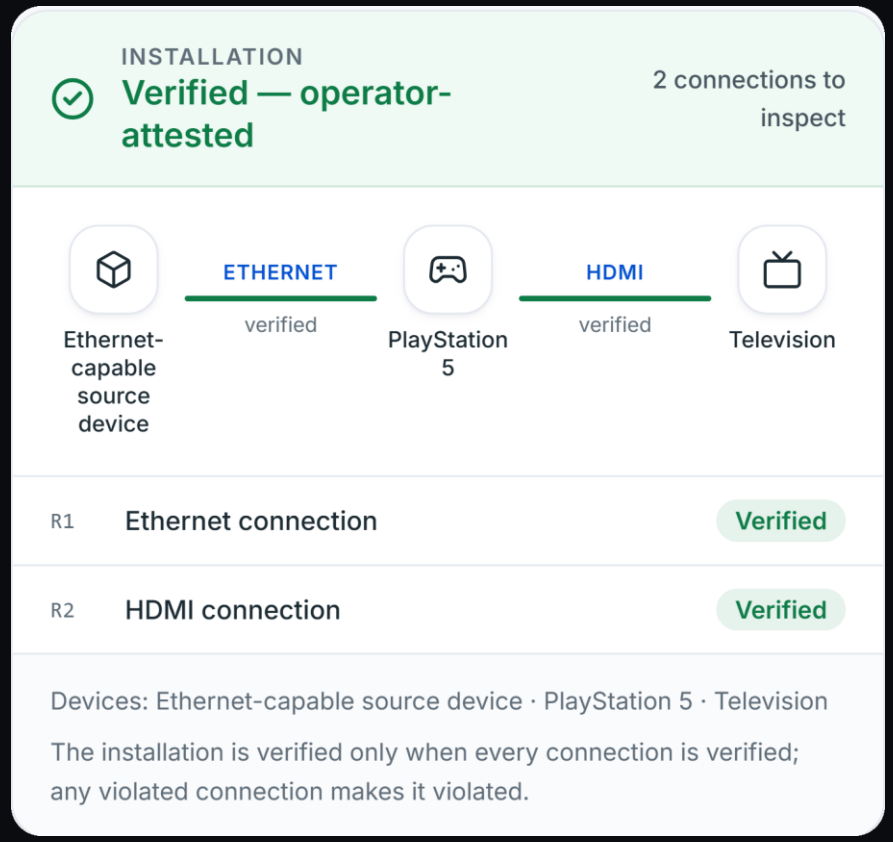
PS5 HDMI OUT plug
attested in writing · Muse saw it, SAM did not corroborate

When visual evidence runs out, ProofCam doesn't invent certainty. It records exactly what the human confirmed.



THE RESULT

Every verdict has an evidence trail.



GENERATED REPORT · EXCERPT (CONDENSED)

```
OVERALL  VERIFIED — OPERATOR-ATTESTED, NOT VISUALLY ESTABLISHED

R1 Ethernet   VERIFIED — OPERATOR-ASSISTED
  photo 2   set-top box RJ45   Muse + SAM (generic) + shape
  photo 4   PS5 LAN            Muse + SAM
  00:13:49  operator: Ethernet run is one cable

R2 HDMI       VERIFIED — OPERATOR-ATTESTED
  photo 3   TV HDMI 1 (ARC)    Muse + SAM (generic) + shape
  00:14:10  operator: PS5 HDMI OUT end attested in writing
  00:14:19  operator: HDMI run is one cable
```

2 / 2 requirements verified

3 of 4 connection ends from photos

3 operator statements, time-stamped

22 live Meta model calls · ≈ $0.13

ENGINEERING

Built for evidence, not just predictions.



- Call budget reserved before every request
- Idempotent, resumable captures
- Contradictions surfaced, never resolved silently
- Operator statements append-only
- 499 backend + 57 frontend automated tests
- Tests use mocked models: regression, not accuracy

≈6,300 lines of Python engine · ≈4,100 lines of TypeScript UI · built 30 Oct – 4 Nov 2026

THE BIGGER OPPORTUNITY

Today, cables. Tomorrow, infrastructure.

PRODUCT VISION · POTENTIAL APPLICATIONS, NOT CURRENT DEPLOYMENTS

Data centres

Is every server patched to the port in the drawing?

Telecom

Is each feed on the right port of the right radio?

Factories

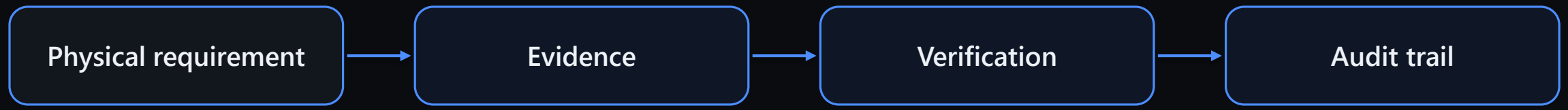
Is the sensor installed and cabled as specified?

Energy

Are string connectors where the design says?

Field service

Was the part replaced, and can we show it?



Same loop. Different specifications.

WHY THIS MATTERS

What if every physical installation could explain why it was trusted?

Verifiable observations

Every claim is tied to a photo and a model output.

Clear uncertainty

Unknown stays unknown. ‘Insufficient evidence’ is an answer.

Traceable decisions

A deterministic policy, not a model's opinion, sets the verdict.

Human accountability

When people vouch, their words are recorded and labelled.

The value isn't naming objects. It's deciding what observable evidence supports a physical requirement.



PROOFCAM

The verification layer for the physical world.

AI that doesn't just see the physical world. It asks for the evidence needed to verify it.

Built with Meta Muse Spark 1.3 and SAM 3.1

github.com/dakshvatss/proofcam

Daksh Vats · Indie Developer Track · Push the Frontier

Independent project, not affiliated with or endorsed by Meta.

What I built during the hackathon

30–31 Oct

M0

Inspection loop, deterministic policy, Meta API client; live Muse + SAM validation on five reference scenarios

2 Nov

M1–M2.2

Resumable agent state, SQLite store, FastAPI service; verdict hardening; next-view guidance and contradictions

3 Nov

M2.3–M3.3

Grounding geometry; mobile camera PWA; private access and spending cap; live-inference readiness

4 Nov

M3.4–M3.9

Three devices / two connections; view relevance; evidence provenance; operator-assisted inspections

4–5 Nov

Demos

Live VIOLATED run (synthetic test photos) and live VERIFIED run on the real home setup

BUILT IN THIS REPOSITORY

- Agent loop: gap selection, next-view requests, attempt ceilings
- Deterministic verification policy and evidence provenance
- Muse/SAM orchestration, grounding geometry, SAM fallback
- SQLite persistence, budgets, idempotent captures, FastAPI
- Mobile PWA, reports, operator statements, test suites

USED, NOT BUILT

- Meta Muse Spark 1.3 and SAM 3.1 via the Meta Model API
- React, Vite, Tailwind CSS, Lucide icons
- FastAPI, Pydantic, SQLite, Pillow, NumPy
- Meta's official SAM output parser (meta_sam_parser)
- All 18 commits dated 30 Oct – 4 Nov 2026 (repository created 30 Oct)

How ProofCam labels what it knows

GROUNDED	Muse claim + SAM mask at the port + spec rules	Counts as evidence
SEEN BY MUSE ONLY	Every check met except SAM corroboration	Recorded, never counted
OPERATOR-CONFIRMED	Plug seated · device identity · one cable (spec-permitted)	Counts; verdict marked operator-assisted
OPERATOR-ATTESTED	A written statement where no photo establishes the end	Counts; verdict marked operator-attested
UNRESOLVED / CONTRADICTED	Missing, ambiguous or conflicting observations	Blocks verification; review if contradictory

Verdicts: VERIFIED · VIOLATED · INSUFFICIENT EVIDENCE — set only by the deterministic policy.

Assurance on a verdict: visual · spec assumption · operator-assisted · operator-attested. Statements never override a contradicting photo.

What ProofCam does not claim

Not certification

Visually observable conditions only: no electrical safety, network connectivity, signal or regulatory claims.

Synthetic data is disclosed

Recording A uses AI-generated test photos. Recording B uses photos of real equipment.

SAM recall is imperfect

In recorded live runs SAM's 'hdmi cable' concept found nothing in 12 of 12 photos; a bounded generic 'cable' fallback is disclosed when used.

Cable identity across photos

Similar cables in different photos are never joined. Without a single-photo trace, identity needs a declared, labelled statement.

Static installations

All photos are assumed to show one unchanged configuration; physical changes must be declared.

Tests ≠ accuracy

Automated tests check policy behaviour with mocked models; no accuracy benchmark has been run.